

Design and Analysis of an Omnidirectional Helical Soft Actuator Based on Fiber Reinforcement and Layer Jamming Effect

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Abstract—Existing helical soft actuators are mostly based on asymmetric pneumatic chambers or unidirectional fiber constraints, resulting in a single deformation direction that struggles to meet the demands of multidirectional adaptive grasping in complex environments. To address this limitation, inspired by the elephant trunk's hydrostatic skeleton and the pangolin's layer jamming mechanism, this paper proposes a novel biomimetic soft actuator (ODHA) integrating fiber reinforcement and variable stiffness characteristics. By independently regulating the pressure of three chambers and external Kevlar fibers, the structure achieves bidirectional, multi-curvature omnidirectional spatial helical coiling. Simultaneously, a built-in PET flap layer utilizes negative pressure to trigger the layer jamming effect, enabling dynamic stiffness modulation. Theoretical models for torsion and bending were established, and the synergistic modulation rules of air pressure and fiber turns on the helical configuration were revealed through simulation and experiments. The results indicate that increasing pressure and fiber turns (e.g., $n=8$) drives the deformation towards highly compact helical coiling, effectively achieving conformal enveloping of irregular targets such as cylinders and boxes. Combined with the variable stiffness structure, ODHA exhibits the synergistic advantage of "soft-conformal to rigid-locking", demonstrating tremendous potential for high-load and highly stable adaptive grasping in unstructured environments, providing novel insights for soft robot design.

Keywords—Soft actuator, Layer jamming, Fiber reinforcement, Helical deformation, Stiffness modulation.

I. INTRODUCTION

Biological tissues such as elephant trunks and octopus tentacles exhibit a typical hydrostatic skeleton structure [1], enabling them to rely on continuous multi-degree-of-freedom spatial deformations to perform complex grasping tasks such as helical coiling. Research on soft actuators based on such biomimetic mechanisms has become a prominent frontier in the field. Benefiting from the rapid development of advanced materials science and 3D printing technology [2][3], the research and development of biomimetic soft robots has significantly accelerated. Compared with traditional rigid robots, soft robots constructed from hyperelastic materials can highly approximate the movement modalities of real organisms, demonstrating significant advantages in system compliance,

adaptability to unstructured environments, and operational safety.

In recent years, scholars have conducted extensive research on soft actuators with helical coiling and grasping capabilities. In terms of designs based on air chamber topology, Gu Guoying's team at Shanghai Jiao Tong University [4][5] developed pneumatic soft actuators using oblique independent elastic air chambers, achieving dexterous winding and grasping of target objects; the research team at the University of Wollongong [6] integrated parallel and oblique air chambers, further enriching the helical movement modalities of the actuators. For underwater operational scenarios, Ji et al. [7] designed a three-dimensional helical deformation soft actuator and successfully applied it to an underwater manipulator; Haitao Qing et al. [8] developed a fully 3D-printed miniature soft hydraulic actuator with a shape memory effect, realizing the non-invasive, multi-modal winding and grasping of small and fragile organisms. Regarding designs based on fiber constraint mechanisms, inspired by the arrangement of muscle fibers in the elephant trunk, Norman Werely's team [9] introduced a single fiber constraint into pneumatic artificial muscles, achieving a series of extensible helical deformations. Majid Shabani et al. [10] developed a fiber-reinforced soft actuator capable of outputting coupled movements of elongation, expansion, and twisting; Ryan Geer et al. [11] achieved a coiling motion that couples twisting, bending, and elongation under internal pressure by winding oblique Kevlar helical fibers around the exterior of a pneumatic actuator. Furthermore, to enhance load-bearing capacity, Saurabh Jadhav et al. [12] proposed a soft robot based on a tunable stiffness mechanism using fiber jamming, whose helical configuration can withstand a load of 330g. Although the aforementioned studies have made significant progress, existing helical soft actuators still have certain limitations in their structures and constraint mechanisms. Most prominently, limited by a single asymmetric air chamber or unidirectional fiber winding arrangement, these actuators can only achieve a single direction of helical movement, making it difficult to meet the demands for omnidirectional adaptation and grasping in complex unstructured environments.

This study proposes an Omnidirectional Helical Soft Actuator that integrates the traditional PneuNets structure, the

three-chamber topology of FMA (Flexible Microactuator), and the fiber reinforcement structure [13]. Using air as the working medium, it achieves omnidirectional helical coiling and grasping capabilities by controlling the pressure magnitude in different air channels. It possesses inherent advantages such as excellent helical coiling movement characteristics, environmental compatibility and adaptability, and low maintenance costs.

II. ACTUATOR DESIGN AND MANUFACTURING

A. Biomimetic Concept

The muscle system of the elephant trunk exhibits typical hydrostatic skeleton characteristics, as shown in Figure 1(b). When performing grasping tasks, its movement manifests as a complex spatial coupling of elongation, bending, and twisting, and it possesses precise output force regulation capabilities to achieve efficient interaction. Inspired by this, this paper designs the internal silicone matrix of the actuator based on the hydrostatic skeleton principle of the elephant trunk. Furthermore, to achieve dynamic stiffness regulation, this paper further draws on the Layer Jamming mechanism of pangolin scales, as shown in Figure 1(c), to design the outer variable stiffness structure, ultimately constructing the ODHA actuator (as shown in Figure 2(d)).

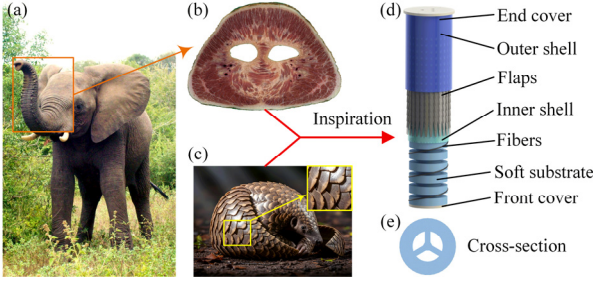


Figure 1. Bio-inspired biomimetic structure of ODHA. (a) Complex spatial movement of the elephant trunk; (b) Cross-section of the muscle arrangement in the elephant trunk; (c) Structure of pangolin scales; (d) ODHA structure; (e) Cross-section of the ODHA matrix.

B. Structural Design

The ODHA matrix is composed of three mutually independent multi-chamber structures, containing one open end and one closed end. The outer layer of the matrix is helically wound with Kevlar fibers; this design integrates PneuNets, the three-chamber topology of FMA (Flexible Microactuator), and the fiber reinforcement mechanism to achieve the pneumatic elongation and bending deformation of the actuator. As shown in the cross-section of the matrix in Figure (e), the three chambers are symmetrically arranged at equal angles along the axis of the actuator matrix. When the three chambers are inflated with equal air pressure, the actuator exhibits pure axial elongation characteristics; when inflated with asymmetric air pressure, the actuator exhibits obvious spatial bending behavior. During this process, the radial constraints provided by the outer rigid rings and fibers strictly limit the lateral expansion of the actuator matrix, thereby efficiently converting the fluid pressure energy into axial or bending deformation. Therefore, the actuator can achieve independent or coupled multi-degree-of-freedom movement outputs of elongation and bending.

A layer of flaps is built-in between the inner and outer shell of the actuator to construct a variable stiffness structure. By applying negative pressure in the sealed interlayer, the friction between the overlapping flaps can be significantly increased, inhibiting their relative sliding. Utilizing this Layer Jamming effect, the system can achieve dynamic regulation of the actuator's stiffness simply by adjusting the magnitude of the negative pressure.

C. Fabrication Procedure

The ODHA matrix is manufactured using lost-wax casting [14], and the process is illustrated in Figure 2. The casting mold is assembled by fastening Model A and Model B, and the Cover for sealing at both ends, using bolts, with the Cavity model precisely positioned at the center of the mold. Silicone is injected into the mold through a pouring hole. After standing to cure for 24 hours, demolding is performed. Utilizing the stiffness provided by the water-soluble core mold before its removal, Fibers are wound and bonded along the helical grooves on the matrix surface using a silicone sealant. Once fibers are securely fixed, matrix is flushed with hot water to dissolve the internal Cavity model, releasing the air chamber. Finally, the Front cover and End cover are sealed at ends, completing fabrication of ODHA actuator.

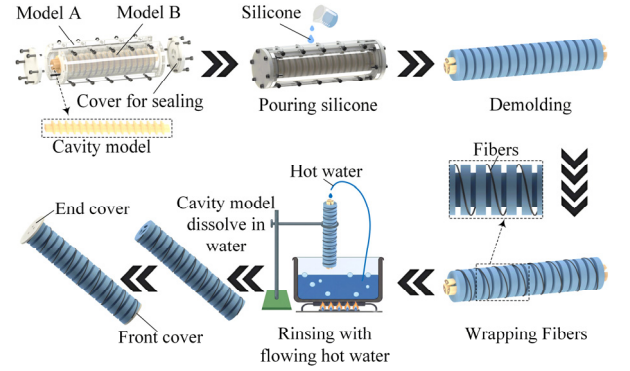


Figure 2. Fabrication process of the ODHA matrix.

The fabrication process of the ODHA variable stiffness structure is illustrated in Figure 3. The PET polyester film is processed by a laser cutting machine and rolled into cylindrical flaps [15]. The exterior of the actuator matrix is sequentially wrapped with an inner Shell, followed by Sewing Flaps, and sealed with an Outer Shell. This process completely encapsulates the flap array between two layers of silicone shell, forming a Layer Jamming variable stiffness structure.

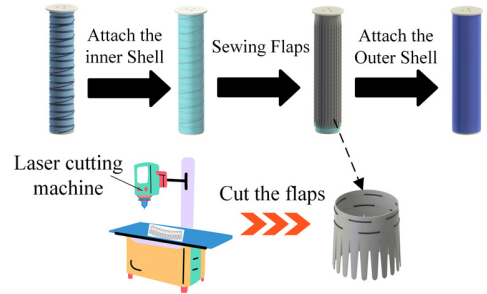


Figure 3. Fabrication process of the ODHA variable stiffness structure.

III. THEORETICAL MODELING

Before establishing the mechanical analytical model of the actuator, to simplify the theoretical analysis and highlight the core movement mechanism, this paper makes the following basic assumptions for the system : (1) the actuator is an ideal homogeneous medium, and its volume is incompressible; (2) the fibers and the soft unit are tightly bonded together, with no relative movement occurring between them; (3) the overall length of the fibers remains constant; (4) given the large length-to-diameter ratio of the designed ODHA actuator, according to Saint-Venant's Principle, the complex non-uniform stress fields induced by end constraints and transition regions are localized at both ends, allowing the central region to approximately satisfy the assumption of a uniform shear stress distribution. (Through multiple experiments, radial deformation has no impact on the grasping of ODHA; in the theoretical modeling, it is only reflected in Equation (10), serving solely for axial bending.)

The shape and parameters of the actuator upon inflation are shown in Figure 6. When a certain amount of compressed air is injected into one of the air chambers of the actuator, due to the asymmetric strain, the actuator bends towards the side with the smaller strain, with a bending angle of θ . Due to the constraint of the Kevlar fibers, the radial expansion deformation of the actuator matrix is hindered, resulting in the axial twisting and elongation of the actuator, with a twisting angle of φ . Let the number of winding turns of the fiber reinforcement layer be n , the length of the actuator be L_0 , the outer diameter be r_0 , the wall thickness be t_0 , A_1 denote the area of the chamber, and A_2 denote the area excluding chamber A_1 . The elongation after inflation is ΔL . Due to the circumferential constraint on the actuator, its outer diameter remains unchanged, and the wall thickness is

$$t = \frac{L_0}{L_0 + \Delta L} t_0.$$

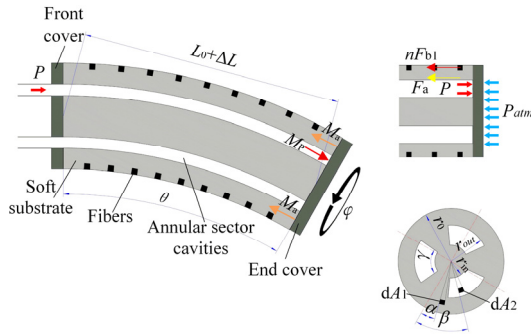


Figure 4. Shape and parameters of the actuator upon inflation.

A. Twisting Model.

Assuming that the force exerted by the fiber reinforcement layer on each actuator matrix is F_b , and the helix angle of the fiber reinforcement layer is α_b , the force can therefore be decomposed into two perpendicular components as shown in Figure 5, where F_{b1} restricts the elongation of the actuator, while F_{b2} induces twisting.

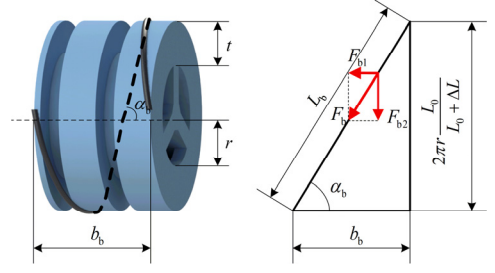


Figure 5. Geometric configuration and force analysis of the actuator matrix.

The two component forces F_{b1} and F_{b2} are given by:

$$\begin{cases} F_{b1} = F_b \cos \alpha_b \\ F_{b2} = F_b \sin \alpha_b \end{cases} \quad (1)$$

The forces inside and outside the actuator satisfy the force balance equation $\Delta P \pi (r_0 - t_0)^2 = n F_{b1} + F_a$, from which F_b can be obtained as:

$$F_b = \frac{\Delta P \pi (r_0 - t_0)^2 - \sigma \pi (2r_0 t - t^2)}{n \cos \alpha_b} \quad (2)$$

Where ΔP is the pressure difference between the inside and outside of the actuator's air chamber; P_{atm} is the atmospheric pressure; and σ is the normal stress of the actuator. The force distribution of F_b along the length direction (in N/mm) can be expressed as:

$$q = \frac{F_b}{L_b} = \frac{\Delta P \pi (r_0 - t_0)^2 - \sigma \pi (2r_0 t - t^2)}{n \cos \alpha_b \sqrt{\left(\frac{L_0}{n}\right)^2 + 4\pi^2 r_0^2}} \quad (3)$$

To calculate the magnitude of the twisting exerted by the fibers of the actuator matrix on the actuator, the actuator unit is divided into infinitesimal circular elements db , obtaining the twisting force of the fibers on the infinitesimal element as $dF_{b2} = dF_b \sin \alpha_b = q \tan \alpha_b db$. The torque generated by the fibers on the infinitesimal element db can then be determined, and the overall torque of the actuator unit can be calculated through integration:

$$dT = r_0 \frac{L_0}{L_0 + \Delta L} dF_{b2} = \frac{r_0 L_0}{L_0 + \Delta L} q \tan \alpha_b db \quad (4)$$

$$\begin{aligned} T_b &= \int_0^{b_b} \frac{r_0 L_0}{L_0 + \Delta L} q \tan \alpha_b db \\ &= \frac{2\pi r_0^2 L_0^2 [\Delta P \pi (r_0 - t_0)^2 - \sigma \pi (2r_0 t - t^2)]}{(L_0 + \Delta L)^3} \end{aligned} \quad (5)$$

Where G is the shear modulus, and I_p is the polar moment of inertia. For a hollow cylinder, $I_p = \frac{\pi}{2} [(r+t)^4 - r^4]$. Substituting this into the basic equation for the relative twisting

angle between the two end cross-sections, $\varphi = \frac{T_b(L_0 + \Delta L)}{nGI_p}$, can obtain:

$$\varphi = \frac{4r_0^2 L_0^2 [\Delta P \pi (r_0 - t_0)^2 - \sigma \pi (2r_0 t - t^2)]}{nG(L_0 + \Delta L)^2 [(r+t)^4 - r^4]} \quad (6)$$

The twisting angle φ is inversely proportional to the number of winding turns n ; furthermore, based on the law of energy conservation, the twisting angle and the elongation of the actuator have a mutually constraining relationship in terms of energy distribution.

B. Bending Model

The moment generated by the applied air pressure on the infinitesimal area A_1 is:

$$dM_p = L_p dF_p = L_p \Delta P dA_1 = L_p \Delta P r dr d\alpha \quad (7)$$

Where L_p is the moment arm of the air pressure, given by

$L_p = \frac{L_0 + \Delta L}{\theta} - r \cos \alpha$, which represents the distance from the line of action of the gas pressure force on the infinitesimal area dA_1 to the axis of rotation; F_p is the axial thrust generated by the air pressure, where $r_{in} \leq r \leq r_{out}$. Let $r_{out}^2 - r_{in}^2 = C$, and $r_{out}^3 - r_{in}^3 = K$. The moment generated by the applied air pressure is:

$$\begin{aligned} M_p &= \int dM_p \\ &= 2 \int_{-\frac{\gamma}{2} - \arcsin\left(\frac{t \cos \frac{\gamma}{2}}{4\gamma}\right)}^{\pi} \int_{r_{in}}^{r_{out}} \Delta P r \left(\frac{L_0 + \Delta L}{\theta} - r \cos \phi \right) dr d\phi \quad (8) \\ &= \frac{\Delta P \gamma (L_0 + \Delta L) C}{2\theta} + \frac{1}{3} K \sin \frac{\gamma}{2} \end{aligned}$$

The elastic force of the actuator matrix on the infinitesimal area A_2 is $dF_a = \sigma dA_2 = E \frac{\Delta L}{L_0 + \Delta L} dA_2$. Based on the geometric relationship, the moment arm of the elastic force can be obtained as $L_a = \frac{L_0 + \Delta L}{\theta} - r \cos \beta$. The moment on dA_2 is $dM_a = L_a dF_a$, and integration yields the moment as:

$$\begin{aligned} M_a &= \int L_a dF_a = \int \sigma d(A_1 + A_2) - \int \sigma dA_1 \\ &= 2 \int_0^{\pi} \int_0^{r_0} E \frac{\Delta L}{L_0 + \Delta L} \left(\frac{L_0 + \Delta L}{\theta} - r \cos \beta \right) r dr d\beta \\ &\quad - 2 \int_{\frac{\pi}{3} - \frac{\gamma}{2}}^{\frac{\pi}{3} + \frac{\gamma}{2}} \int_{r_{in}}^{r_{out}} E \frac{\Delta L}{L_0 + \Delta L} \left(\frac{L_0 + \Delta L}{\theta} - r \cos \alpha \right) r dr d\alpha \quad (9) \\ &\quad - 2 \int_{\frac{\pi}{2}}^{\pi} \int_{r_{in}}^{r_{out}} E \frac{\Delta L}{L_0 + \Delta L} \left(\frac{L_0 + \Delta L}{\theta} - r \cos \alpha \right) r dr d\alpha \\ &= E \frac{\Delta L}{2\theta} [2\pi r_0^2 - 3C\gamma] \end{aligned}$$

From Figure 6, the moment equilibrium $M_p = M_a$ can be obtained. Combining Equation (8) and Equation (9) yields:

$$\theta = \frac{3E\Delta L [2\pi r_0^2 - 3C\gamma] - 3\Delta P \gamma (L_0 + \Delta L) C}{2K \sin \frac{\gamma}{2}} \quad (10)$$

The bending angle θ of the actuator is inversely proportional to the gas pressure in the inner chamber, that is, the greater the gas pressure in the inner chamber, the smaller the bending angle θ of the actuator, meaning the greater the degree of bending of the actuator.

C. Error Analysis

To verify the accuracy of the established analytical mechanical model, an error analysis was conducted on the bending and twisting deformations of the actuator. Under the same air pressure input conditions, 100 sets of bending and twisting angles from theoretical calculations and simulation experiments were extracted respectively for comparative evaluation, as shown in Figure 6.

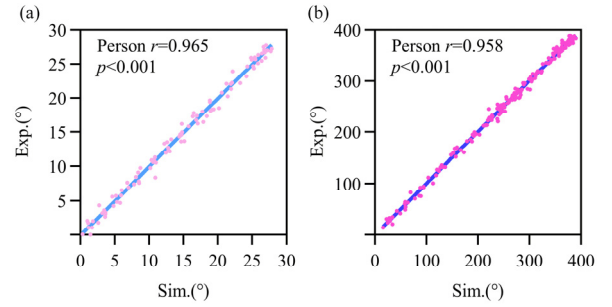


Figure 6. Scatter plot of theoretical and simulated bending and twisting angles.

Through quantitative statistical analysis using the Pearson correlation coefficient, the results show that: in terms of bending deformation, the theoretical bending angle and the simulated bending angle exhibit a highly significant positive correlation ($r=0.965$, $p<0.001$); in terms of twisting deformation, the two results also demonstrate a high degree of consistency ($r=0.958$, $p<0.001$). The extremely small p-value further proves the statistical significance of this correlation. This indicates that the theoretical analytical model established in this paper can effectively and accurately predict the spatial axial twisting and bending behaviors of the actuator under different internal chamber pressures.

IV. PERFORMANCE EVALUATION AND EXPERIMENTAL ANALYSIS

A. Deformation Simulation

To investigate the three-dimensional helical coupled deformation mechanism of ODHA under single-chamber actuation, a simulation analysis is conducted on the actuator. The material parameters used in the simulation are shown in Table 1. The actuator matrix is meshed using solid elements, while the fibers are modeled using cable elements. The contacts among the actuator matrix, the fibers, and the top and bottom covers are all defined with tie constraints.

TABLE I. MATERIAL PROPERTIES FOR FINITE ELEMENT SIMULATION

Materials	Model	Parameters	Values
Silicone	Yeoh	$C_{10}(\text{MPa})$	0.11
		$C_{20}(\text{MPa})$	0.02
		$C_{30}(\text{MPa})$	0
Kevlar fiber	Elastic	$E(\text{MPa})$	31067
		ν	0.36
9100E	Elastic	$E(\text{MPa})$	2600
		ν	0.42

The simulation analysis of twisting and bending corresponding to different numbers of turns at 0-0.1 MPa with a gradient of 0.02 MPa is shown in Figure 7. The simulation results intuitively demonstrate that the bending moment generated by the eccentric air chamber and the radial/axial constraints provided by the outer fibers produce a significant coupling effect, successfully inducing the three-dimensional helical motion of the actuator. As the number of fiber winding turns n increases, the restrictive effect of the fiber reinforcement layer on the radial expansion of the matrix is enhanced, promoting the conversion of more air pressure energy into axial twisting strain energy. Therefore, the spatial configuration of the actuator gradually evolves from "large-curvature spatial bending" at a low number of turns (e.g., $n=1, 2$) to "highly compact helical winding" at a high number of turns (e.g., $n=4, 8$). Specifically, when $n=8$, the cumulative twisting angle at the end of the actuator reaches 385° , demonstrating excellent potential for omnidirectional spatial wrapping. (The 385° represents the maximum bending angle obtained from the simulation at an input pressure of 0.1 MPa, and this angle can also be measured in experiments. It serves solely as a peak value illustration to demonstrate the bending capability of the actuator.)

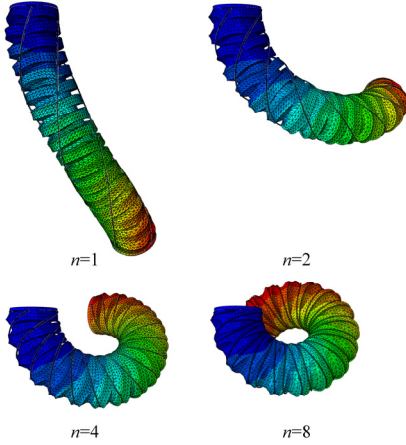


Figure 7. Total displacement magnitude of ODHA under different numbers of turns after inflation with 0.1 MPa.

B. Spatial Trajectory Analysis

To further quantitatively evaluate the spatial movement performance of ODHA, the experiment recorded the three-dimensional deformation data of the actuator under increasing air pressure and different numbers of fiber turns. Figure 8 and Figure 9 extract and plot the displacement evolution trajectories of the actuator's central axis and end point in the three-dimensional space (x, y, z), respectively.

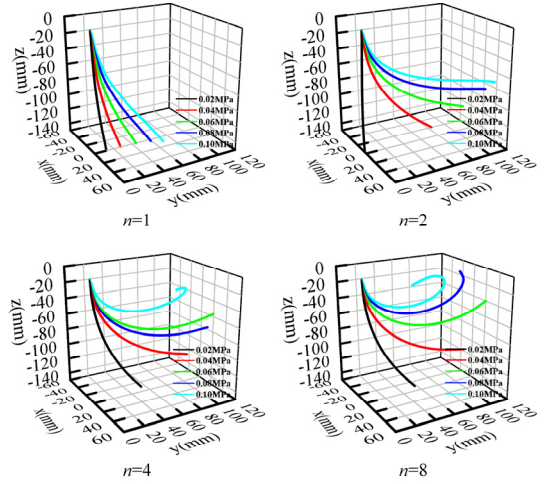


Figure 8. Three-dimensional spatial trajectories of the ODHA central axis varying with air pressure under different numbers of fiber turns.

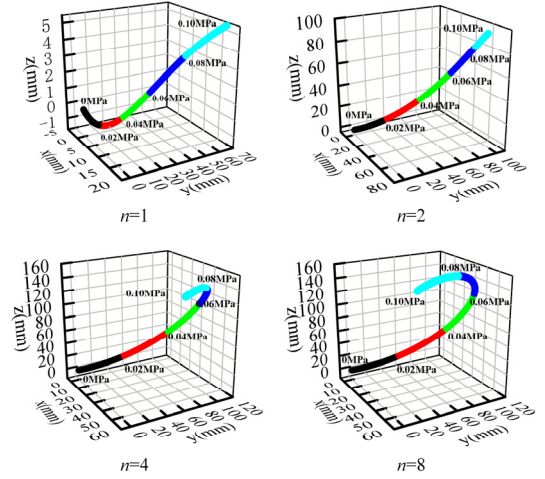


Figure 9. Three-dimensional spatial trajectories of the actuator's end point varying with air pressure under different numbers of fiber turns.

The experimental curves reveal the synergistic modulation rules of input air pressure and fiber constraints on the spatial configuration. On the one hand, with the monotonic increase of air pressure, the end point of the actuator traces a continuous helical extension trajectory in space; on the other hand, the number of fiber turns n determines the topological characteristics of the helical enveloping space. Specifically, under the condition of a low number of turns ($n=1$), the end trajectory mainly manifests as large-span bending in a single direction; whereas under the condition of a high number of turns ($n=8$), the end trajectory exhibits a significant inward contraction trend in the x - y projection plane, accompanied by elongation along the z -axis. This high-curvature helical trajectory can effectively enhance the conformal enveloping area of the actuator for complex target objects.

C. Adaptive Grasping Experiment

To intuitively verify the omnidirectional adaptive grasping capability of ODHA in actual unstructured environments, this paper conducted a series of grasping experiments targeting

objects with various shapes, sizes, and surface characteristics (as shown in Figure 10). Figure 10(a) illustrates the initial vertical compliant state of the actuator without inflation. When compressed air is injected into a specific air chamber, ODHA, relying on its excellent helical coiling characteristics, can quickly conform to the outer contours of the target objects for dynamic winding.

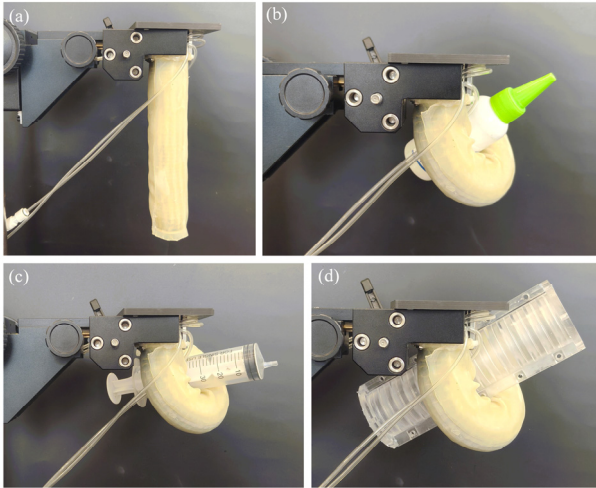


Figure 10. Grasping capability experiment. (a) Natural state of ODHA; (b) Grasping a cylindrical paint bottle; (c) Grasping a cylindrical syringe; (d) Grasping an irregular mold.

To intuitively verify the omnidirectional adaptive grasping capability of ODHA in actual unstructured environments, a series of grasping experiments were conducted on objects with various shapes and sizes (as shown in Figure 10). When a specific chamber is inflated, ODHA leverages its helical coiling characteristics to quickly conform to target contours. Specifically, for cylindrical bottles and syringes (Figures 10(b)-(c)), the soft silicone matrix forms large-area contact to deliver gentle grasping forces without damaging fragile targets, whereas for irregular molds with right-angled edges and complex textures (Figure 10(d)), the actuator utilizes high-curvature three-dimensional trajectories guided by high fiber turns to achieve highly fitted conformal enveloping.

More crucially, ODHA exhibits an exceptional "soft-rigid transition" advantage during operation. Upon completing the initial compliant enveloping, activating the Layer Jamming effect via vacuum negative pressure significantly increases friction between the overlapping PET flaps. This innovative "initial compliant fitting, subsequent rigid locking" mechanism not only tightly anchors the actuator to the target surface but also substantially enhances overall anti-bending and anti-twisting stiffness, effectively overcoming the physical bottlenecks of weak load capacity and poor stability in traditional soft actuators, and verifying its high-load, stable adaptive grasping potential in complex environments.

V. CONCLUSION

Inspired by the elephant trunk and the pangolin, this paper develops a biomimetic soft actuator (ODHA) integrating PneuNets, FMA topology, and Kevlar fibers, achieving spatial multi-degree-of-freedom adaptive bending and twisting.

Theoretical models and experiments verify the synergistic regulatory effects of air pressure and fiber turns (e.g., $n=8$) on the helical configuration, enabling highly fitted conformal enveloping of complex geometries. Furthermore, the layer jamming effect provides an excellent "soft-rigid transition" characteristic during grasping, significantly enhancing the system's structural stiffness and anti-slip capability, which comprehensively overcomes the physical bottlenecks of weak load capacity and poor stability inherent in traditional soft actuators. Future research will focus on expanding the axial length and fiber turns of the actuator, introducing closed-loop sensing and compliant control algorithms, and exploring multi-material integrated 3D printing manufacturing processes to further advance the in-depth application and industrialization of this biomimetic soft actuator in unstructured environments.

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